

Solution Requirements

Date	1 October 2026
Team ID	SWTID-2026-3594
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users sign in to the ServiceNow instance with their own credentials; the administrator impersonates the EEE User to verify access.	High
2	Authorization levels	Custom roles bb1 (read), bb2 (create), bb3 (write) and bb4 (delete) are assigned to the EEE User; the admin role keeps full access to all records.	High
3	External interfaces	None. The solution is built entirely on native ServiceNow features (Users, Roles, Tables, ACLs) with no third-party APIs.	Low
4	Transactions processing	Every read, create, write and delete request on u_institution_details is evaluated by the matching ACL before it is allowed.	High
5	Reporting	The Institution Details list shows only the records the user is allowed to read; list views are used to verify results for each user.	Medium
6	Business rules, etc.	Only users with bb1 can read records whose Branch is EEE; Create needs bb2, Write needs bb3, Delete needs bb4; admins bypass the script.	High
7	Compliance to laws or regulations	Follows the principle of least privilege and protects student personal data (name, email, phone) from unauthorized viewing.	Medium
8	Other	Student Roll Number is generated as an Auto Number; the security_admin role must be elevated to create ACLs.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	ACL scripts must be lightweight (role checks only, no extra database queries) so lists and forms load quickly.	List loads in < 2 seconds with ACLs active
2	Scalability	The same ACL pattern works for any number of records and users without changing the script.	Works for all records of u_institution_details
3	Security & Data Privacy	Access is denied by default; each operation needs its own role, and the condition is enforced on the server side.	User without bb1 sees 0 records
4	Reliability & Availability	Runs on the ServiceNow cloud platform; ACL rules persist and apply on every request.	ACLs active for 100% of requests
5	Usability & Accessibility	Users work in the standard ServiceNow UI; the New button appears only for users with the create role.	No extra training needed for users
6	Other	Maintainability: script is short, commented and uses consistent role names bb1 - bb4.	Script readable in under 5 minutes